

Quality Communication using Varying Parameters for Wireless Temporary Ad hoc Networks

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ABSTRACT

A little packet known as a beacon packet is transmitted; it signals the existence of a base station, and mobile units detect the beacons and make an effort to connect wirelessly. Geographic routing methods have gained a lot of attention recently because they solely base their forwarding decisions on the local topology. MANET nodes can be identified by their limited resources, which include storage, bandwidth, remaining energy, routing coordination, end-to-end latency, and multi-path communication. Because of topology, MANET routing is a significant problem. This work builds upon my previously published paper and proposes multipath work with three distinct topology sizes, three distinct node sets, and three distinct malicious node sets. It uses Network Simulator 2 (NS2) to simulate a variety of parameters, including packet delivery ratio, throughput, routing overhead, packet loss, delay, and remaining energy.

Keywords:-MANET , routing, topology, distinct, packet

INTRODUCTION

Because the network topology in a MANET is dynamic in nature, any node can join or depart at any time. Each node acts as both a host and a route in an autonomous manner. Since all nodes have the same characteristics, responsibilities, and abilities, they create a perfectly symmetric environment. Mobile nodes are distinguished by their low memory, low power consumption, and lightweight designs.

To share information between devices in a non-fixed network, the mobile ad hoc router model essentially designed three different sorts of connections: pre-established connections for proactive, on-demand connections established for reactive, and both concepts available in hybrid routing protocols. Extension of research is being conducted in recent years in multipath fading communication based

on some criteria like minimum cost, minimum weight, maximum forwarding capability and maximum receiving capability, etc. Multipath importance techniques use alternative multiple paths in networks which can elide provide such as tolerance increase bandwidth and improving security.

PROBLEM IDENTIFICATION

The broadcasting nature of radio transmission causes transmission congestion, which forces the routing protocols to update the routing tables frequently due to the dynamic nature of MANETs. A packet may need to be routed through intermediate nodes in order to reach its destination since nodes in a MANET are unable to communicate directly with nodes outside of their communication range. Monitoring the limitations in intermediary nodes (multi-

hop routing) also becomes crucial. As a result, route failures could result from an effective routing strategy. Finding a path free of rogue nodes is the most straightforward MANET routing strategy. The objective of this research is to offer an unbreakable path to enhanced residual energy for a secure transmission method.

As a result, Ad hoc on-demand Distance Vector with Help of Fuzzy Controller (FC-AODV), a new routing technique, is proposed.

Simulation Parameter: The simulation parameters are listed below, and nodes are initially placed at predefined positions.

Table 1:-Simulation parameters

Parameter	Values
Simulation area	780m*780m
Number of nodes	50, 100, 150
Number of packets sender	25
Constant bit rate	4 (packets/second)
Packet size	512 bytes
Initial energy/node	100 joules
Antenna model	Omni directional
Simulation time	500 sec
Malicious node	8

RESULT AND DISCUSSION

In this section, we analyzed the findings and talked about the suggested and current

approaches for classifying three distinct models using NS 2.

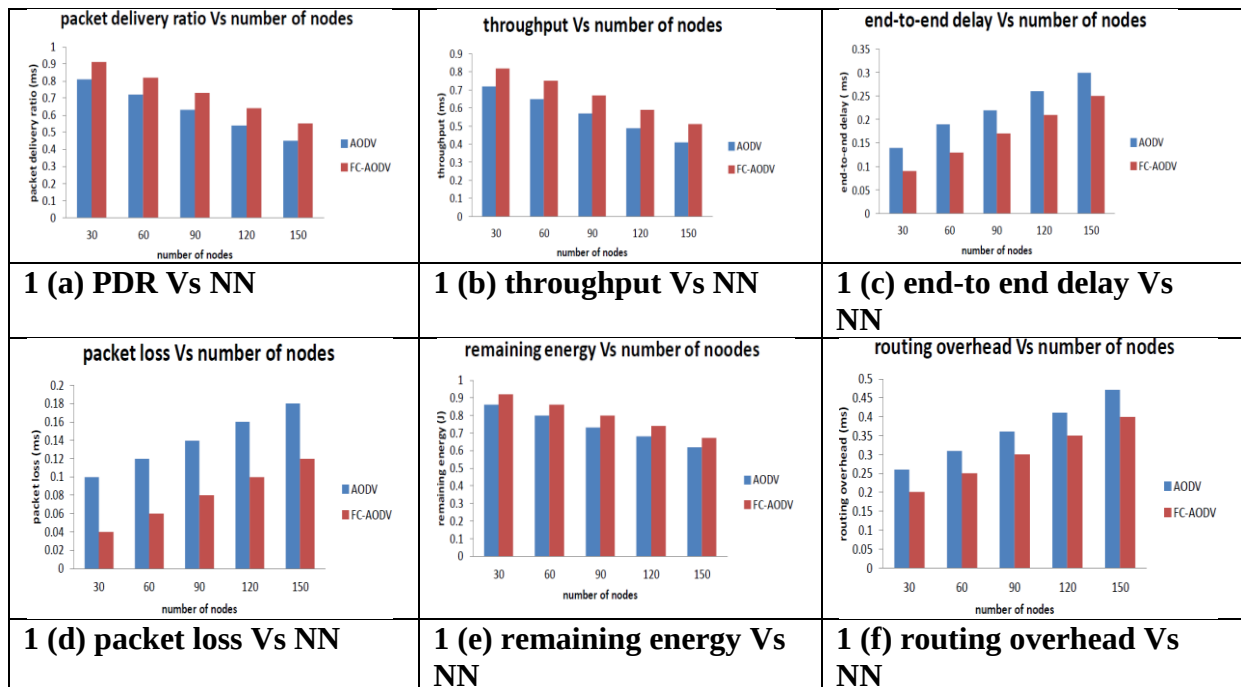


Fig.1:-Model : Results of Parameter Values (SA=780m, NN=150 & MN=8)

The outcome is displayed in Figure 1. The values of (a), (b), (c), (d), (e), and (f) are changed from 1 to 8 out of 150, and

simulation is done using both suggested and current methods to determine the packet delivery ratio. The simulation

results shown in Fig. 1 demonstrate that our proposed model increases average throughput by 10%, minimizes average packet delivery ratio by 10%, reduces average delay by 6.2%, lowers average losses by 6%, increases average remaining energy by 6%, and reduces average routing overhead by 6.2% when compared to AODV.

CONCLUSION

This manuscript makes it abundantly evident that, if wireless communication is conducted without acknowledgement, misbehavior attacks for multi-path routing have always posed a serious threat to the security of MANETs during transmission drop (or) attack the packet. This study proposes the FC-AODV routing protocol, which uses a fuzzy controller.

Through network simulation 2, the simulation results compare the proposed FC-AODV algorithm with the current AODV algorithm in various circumstances. In comparison to the current routing protocol, the developed model's ability to identify misbehaving nodes improves average packet delivery ratio for all three scenarios by 12.4%, average throughput for three scenarios by 10.27%, average end-to-end delay for all three scenarios by 5.37%, and average packet loss for all three scenarios by 5.57%. additionally address the shortcomings of the current approach.

The results of all the scenarios unequivocally demonstrate that the proposed system increased average remaining energy by 6.4% over the current method and decreased average routing overhead by 6.37 over the current routing protocol. In our upcoming research, we intend to look into the following problems. 1) The same idea can be used with satellites to conserve more energy and lessen traffic on the trip. 2) It is possible to investigate the options for using the

shortest path algorithm in place of the redistribution requirement. 3) Rather of using software simulation, the performance of FC-AODV can be evaluated in a real-time network environment.

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